

Summary of the project:

Nowadays society faces a serious challenge due to the rapidly increasing number of chemicals used in everyday life and industry. The European chemicals regulation **REACH** (Registration, Evaluation, Authorisation and Restriction of Chemicals), adopted in 2007, imposes strict requirements to enhance the protection of human health and the environment from chemical risks. Traditional laboratory animal testing is time-consuming, costly, and raises ethical concerns. As a result, REACH encourages the use of **alternative methods** for toxicity assessment. This shift is also reflected in other European regulations, such as the **Cosmetics Regulation**, which imposes a complete ban on animal testing of cosmetic products and their ingredients.

The alternative approaches include **in vitro** and **in silico** methods, with the latter offering significant advantages in terms of speed and cost. Over the recent decades this has led to growing interest in the rapidly evolving field of **computational toxicology**. The development of freely accessible **in silico models** and their integration into software tools or web-based platforms is particularly important for researchers in the fields of computational chemistry, biology, and toxicology as well as for regulatory authorities and industry professionals seeking efficient solutions for risk assessment and substance prioritization.

To address these needs, the current project builds upon existing *in silico* models for assessing toxicokinetic and toxicodynamic parameters of substances. It aims to expand the model base with additional validated models from scientific literature and to develop new ones relevant to regulatory toxicology. These models will be integrated into an open access web-based platform, CompuTox Predictor, offering broader functionality compared to existing solutions. The platform will include both QSAR (Quantitative Structure–Activity Relationship) models and structure-based molecular modeling tools, enabling comprehensive toxicity assessments.

The project ultimately aims to support **Bulgarian regulatory authorities and industry** in evaluating chemical risks in compliance with the REACH regulation.

The project includes the following research work packages:

Work Package 1: “Development of a Database of In Silico Models and Supporting Data for the Assessment of Toxic Effects on Human Health”

This work package focuses on building a **database of *in silico* models for assessing toxic effects on human health**, alongside a **repository of chemical compounds** containing structural information, molecular descriptors, and experimental toxicity data.

Core activities include the integration and validation of existing QSAR models — among them, **an in-house model for predicting membrane permeability** (PAMPA – Parallel Artificial Membrane Permeability Assay) and gastrointestinal absorption. The project will also identify **publicly available QSAR models and will develop new QSAR models for the assessment of toxicokinetic and toxicodynamic properties** through the selection and analysis of appropriate structural and experimental data. Additionally, **structure-based models for prediction of ligands of nuclear receptors such as ER α and PPAR γ** will be developed using molecular docking techniques. All models will be documented in the standardized **QMRF format** (QSAR Model Reporting Format). Both the structured model database and the centralized repository of chemical and toxicological data will be

implemented in the web-based platform **CompuTox Predictor**.

Work Package 2: “Development of a Database of In Silico Models and Supporting Data for the Assessment of Toxic Effects on the Environment”

Work Package 2 focuses on the development of a database of *in silico* models and associated data for the assessment of **toxic effects on the environment**. The primary goal is to build a structured and functional modeling system to support regulatory evaluation of the ecotoxicity of chemical substances.

The work package will include existing models for **acute toxicity** for various aquatic organisms, including crustaceans (*Daphnia magna*), algae (*Pseudokirchneriella subcapitata* and *Scenedesmus subspicatus*), and fish (*Brachydanio rerio* and *Oncorhynchus mykiss*). All models will be presented in the **QMRF format** (QSAR Model Reporting Format) and incorporated into a unified database. Additionally, correlation models will be integrated to enable prediction of toxicity to one species based on experimental toxicity data from another species.

The second major activity focuses on **the development of new in silico models**, expanding the range of studied biological effects and species. For this purpose, chemical substances with known structures and experimental data will be selected through a literature review. These compounds will be characterised using appropriate **structural descriptors** and subsequently employed in the construction of new models. The developed models will undergo validation in accordance with OECD principles to ensure their applicability within regulatory frameworks.

Another major activity is **the assessment of the toxicity of chemical mixtures** using established models: the Concentration Addition (CA) model, Independent Action (IA) model, Generalized Concentration Addition (GCA) model, QSAR-TSP (QSAR two-stage prediction) modeling, and others.

The ultimate goal is to create a comprehensive **database of chemical substances**, including their structures, structural descriptors, and experimental data on environmental toxicity. The models themselves will be implemented in the web-based platform **CompuTox Predictor**, facilitating regulatory and scientific work in ecotoxicology.

Work Package 3: “Development of a Web-Based Software Platform Integrating the Developed Model Databases for Predicting Toxic Effects of Chemical Substances”

Work Package 3 focuses on the development of an **open-access web-based platform**, in which the *in silico* models selected/developed in WP1 and WP2, as well as the chemical structure and experimental databases accumulated in these work packages will be implemented.

The first step of development of the web-based platform is selection, installation and setup of the newly purchased hardware, including the operating system, development environment and application software of the new cluster, setup of the platform's web server, as well as a setup of a secure communication protocol between nodes of the web-based platform. The next step involves the implementation of the developed/selected in WP1 and WP2 models in the web-based platform. The models' backend location on specific node(s) is selected depending on the required computing resources. In addition, the input-output and the presentation models for data are developed and implemented. The next stage involves the construction of the databases of compounds used in model development employing an appropriate database management system.

The selected models will be available to the users through a **two-tier web-based platform**. The first tier will be a web server, through which the users will submit their queries in the form of chemical structures (either individually or as a list, depending on the computational time required to generate the results) and receive the calculated predictions along with relevant data for the selected model (calculated structural descriptors, model parameters, confidence levels of the predicted results, 3D coordinates of computed docking poses, affinity scores, etc.).

For models that do not require significant computational resources, predictions will be generated at the first tier of the platform. For more computationally demanding models, the second tier—a mini-cluster capable of multithreaded calculations—will be activated. The calculations can be executed on a single cluster node or in parallel across multiple cluster nodes.

The cluster's software infrastructure will include **AutoDock Vina** (molecular docking), **OpenBabel** and **RDKit** (molecular formats manipulation according to model requirements), **PaDEL** (structural descriptors calculation), **Python** (statistical modeling), etc.